MIL-R-24085A(SHIPS)
AMENDMENT-1
17 October 1975

MILITARY SPECIFICATION

REFRIGERANT UNIT, CENTRIFUGAL FOR AIR CONDITIONING

This amendment forms a part of Military Specification MIL-R-24085A(SHIPS), dated 18 March 1968 and is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1 and 2

2.1, under "SPECIFICATIONS FEDERAL": Add:

"PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall."

2.1, under "SPECIFICATIONS MILITARY": Delete reference to "MIL-V-3" and "MIL-H-15424" and add:

"MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-B-22191 - Barrier Materials, Transparent, Flexible, Heat-Sealable."

2.1, under "STANDARDS MILITARY": Delete reference to "MIL-STD-167" and "MIL-STD-758" and add:

"MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-721 - Definitions of Effectiveness Terms for Reliability Maintainability, Human Factors, and Safety.
MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods."

PAGE 2 and 3

2.2: Delete "AMERICAN SOCIETY FOR TESTING AND MATERIALS" Standards and "UNITED STATES OF AMERICA STANDARDS INSTITUTE" standard and substitute:

FSC 4130

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"AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A 105 - Forgings, Carbon Steel, for Piping Components.
A 204 - Pressure Vessel Plates, Alloy Steel, Molybdenum.
A 266 - Carbon Steel Forgings for Seamless Drums, Heads, and other Pressure Vessel Components.
A 515 - Pressure Vessel Plates, Carbon Steel, for Intermediate- and Higher-Temperature Service.
A 536 - Ductile Iron Castings.
B 61 - Steam or Valve Bronze Castings.
B 98 - Copper-Silicon Alloy Rod, Bar, and Shapes.
B 139 - Phosphor Bronze Rod, Bar, and Shapes.
B 169 - Aluminum Bronze Plate, Sheet, Strip, and Rolled Bar.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

"AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
B 46.1 - Surface Texture.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)"

2.2: Add:

"ELECTRONIC INDUSTRIES ASSOCIATION (EIA)
Safety Engineering Bulletin No. 3 - System Safety Analytical Techniques.

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)"

PAGE 5

Add as paragraph 3.3.4:

"3.3.4 Hazard analysis. Hazard analysis shall be performed to identify hazardous conditions for the purpose of their elimination or control. Analysis shall be made to examine the equipment, subsystems, components, and their interrelationship to training, maintenance, and operational environment. EIA bulletin No. 3 may be used as guidance for selection and use of safety analysis techniques."

3.4.3, line 1: Delete "One purge unit" and substitute "One purge and pump-out unit."

PAGE 9

3.5 through 3.5.2: Delete and substitute:

"3.5 Reliability assurances.

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"3.5.1 The principle of maximum reliability is paramount and no compromise of this principle shall be made with any other basic requirement of design. Where wear or corrosion is unavoidable, the parts subjected to such wear or corrosion shall be of the best material for the purpose in order to reduce those detrimental effects to a minimum.

"3.5.1.1 The air conditioning unit and its components shall be of such design that they will have an operating life of at least 60,000 hours when subjected to, at various times, any or all of the shipboard conditions specified.

"3.5.2 Definitions. Reliability terms shall be defined in accordance with MIL-STD-721 except as otherwise defined in this specification.

"3.5.3 Reliability requirement. The minimum acceptable mean time between failure (MTBF) for each air conditioning unit shall be 15,000 operating hours.

"3.5.4 Air conditioning unit failure. A failure of an air conditioning unit shall be defined as any event which requires corrective maintenance to be performed on the unit.

"3.5.5 Reliability program plan. The supplier shall submit a reliability program plan for review. The supplier's reliability program plan shall describe how he proposes to conduct his reliability program to meet the requirements of 3.5.3. The supplier's program plan shall incorporate, as a minimum, the reliability program elements of MIL-STD-785 specified in 3.5.7.

"3.5.6 Reliability analysis report. The supplier shall submit a reliability analysis report. The report shall consist of data which the supplier has recorded on centrifugal air conditioning units he manufactures. The data shall have been collected over a 3-year period during actual field operating experience exclusive of prototype testing. The reliability analysis report shall list the following data:

(a) Unit serial numbers.
(b) Unit model numbers.
(c) Quantity in service.
(d) Service time for each unit.
(e) Number and description of failures.

"3.5.7 Reliability program elements. A reliability program shall be planned and followed by the supplier utilizing those procedures of MIL-STD-785 necessary to achieve the specified MTBF. This reliability program shall consist of at least the following elements of MIL-STD-785 (see MIL-STD-785 for a detailed explanation of each element):

(a) Subcontractor and supplier reliability programs.
(b) Design techniques.
(c) Reliability analysis.
(d) Parts reliability.
(e) Failure mode and effect analysis.
(f) Reliability critical items.

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(g) Design reviews. The procuring activity shall be notified at least 15 days prior to each design review to permit possible participation by the procuring activity.
(h) Transition from development."

PAGE 10

3.12: Delete and substitute:

"3.12 Noise.

"3.12.1 Airborne. Type I and type II units shall be designed and tested for airborne noise as specified in 4.4.1.1.

"3.12.2 Structureborne. Type I units, as specified (see 6.1), and type II units shall be designed and tested for structureborne noise as specified in 4.4.1.2."

3.13.3: Delete and substitute:

"3.13.3 The units shall be designed and tested to meet the requirements of MIL-STD-167-1, type I environmental vibration from 4 to 33 hertz (Hz), as specified in 4.4.1.3."

PAGE 11

3.15.3.1: Add "Sight glasses shall be provided in the compressor oil reservoir or crankcase to permit observation of the normal operating and shut down oil levels. The oil heaters shall be replaceable without disassembly of the compressor."

PAGE 12

3.16.3: Delete "(b) Interlock with water pumps."

3.16.3.1 and 3.16.3.2: Delete and substitute:

"3.16.3.1 Controls shall be provided so that the compressor motor will not start unless there is sufficient chilled water flow through the chiller and will stop if the chilled water flow through the chiller falls below an acceptable level. Safety devices shall be provided so that the compressor motor cannot be started without adequate condenser water supply and will stop when the condenser water supply is insufficient."

PAGE 13

3.18.1.1: Delete and substitute:

"3.18.1.1 Provisions for installation of zinc anodes in accordance with MIL-A-19521 shall be made in sea water condensers. The zincs shall be capable of being replaced without removing the heads."

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PAGE 16

Add as paragraph 3.21.2.1:

"3.21.2.1 The chiller tube sheets shall be machined and grooved to accommodate the chiller tubes. The inlet and outlet ends of the tubes shall be expanded into tube sheets by an automatic tube expander. Tube holes on the water side of the tube sheet shall be deburred and finished to a 1/16-inch 45-degree chamfer to remove the corner. The inlet end of the tube shall be belled and finished flush with the face of the tube sheet."

3.21.3: Delete and substitute:

"3.21.3 Provision shall be made for purging air and draining water from the chiller heads. Bolts and nuts for fastening the heads shall be aluminum-bronze in accordance with ASTM B 169, phosphor-bronze in accordance with ASTM B 139, or copper-silicone alloy in accordance with ASTM B 98. All joints on the chiller shall be silver brazed or welded. The chiller shall be insulated in accordance with MIL-STD-769. Lifting lugs or eyes shall be provided to facilitate head removal."

PAGE 17

3.24 Make the following changes:

Item (f): Delete and substitute: "(f) Digital elapsed time indicator to show hours of compressor operation."
Item (i) (3): Add "or pressure."
Item (i) (6): Add "or a chilled water flow switch which meets the applicable tests of MIL-S-16032 and is adequately sized for the application."

PAGE 18

3.24.2.1.1: Delete and substitute:

"3.24.2.1.1 The sensing line for each pressure indicator or pressure switch on the air conditioning unit or purge pump-out unit shall contain a root valve to minimize refrigerant leakage if the sensing line is broken. In addition, the sensing line for each pressure indicator or switch shall contain a gage valve and test connection to isolate the indicator or switch for repair and to check for accuracy using an external pressure source without removing the pressure indicator or switch from the control panel. Wherever possible, pressure indicators and switches which monitor the same pressure shall use a common sensing line."

3.24.2.2: Add:

"(g) Condenser liquid refrigerant outlet-range of minus 40°F to 180°F."

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PAGE 18 and 19

3.26 through 3.26.5: Delete and substitute:

"3.26 Water regulating valve. An automatic water regulating valve shall be provided to maintain sufficient condenser pressure and stable operation of the air conditioning unit for a sea water inlet temperature range specified in 3.1.1. The valve shall be designed to operate using condenser refrigerant pressure to vary the sea water flow to the unit. The valve shall be designed to withstand the pressure specified in 3.19. The valve will be installed in the condenser sea water discharge piping. The valve shall be designed for a maximum allowable pressure drop of 15 lb/in^2. An override shall be provided to permit manual operation in the event of valve failure. Flow through the valve shall be from under the seat tending to open the valve.

"3.26.1 The valve shall be constructed of nonferrous or corrosion-resisting material suitable for sea water. The valve body shall be made of valve bronze, ASTM B 61; cast 70-30 copper-nickel alloy, MIL-C-20159; or forged 70-30 copper-nickel alloy, MIL-C-15726. The internal parts of the valve which come in contact with sea water shall be constructed of nickel-copper alloy, QQ-N-286 or QQ-N-281.

"3.26.2 The sensing line from the air conditioning unit to the water regulating valve shall be provided with a root valve so the line can be isolated if it is damaged.

"3.26.3 The water regulating valve shall be constructed such that sea water cannot enter the refrigerant circuit during normal operations or in case of derangement.

"3.26.4 The valve and actuator shall be capable of passing the shock requirements of MIL-S-901."

3.28, last sentence: Delete and substitute "An electronic thermopile vacuum indicator with gage tube shall be supplied. The indicator and tube shall be capable of accurately reading a vacuum of 20 millimeters (mm) of mercury down to at least 500 micrometres. An adapter for this application shall be provided for attachment to the air conditioning unit."

3.28.4 Delete and substitute:

"3.28.4 The electronic vacuum indicator shall be provided complete with portable storage box to protect it from accidental damage and for storage of electrical cord, gage tube, and adapters."

3.29, lines 1 and 2: Delete "for type II installations."

3.30: Delete and substitute:

"3.30 The air conditioning unit and purge pump-out equipment shall be designed such that contamination of the ship's atmosphere with refrigerant vapor is minimized during operation, maintenance, and repair."

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PAGE 22

3.34: Add "Existing drawings which have been reviewed and found satisfactory, where applicable, may be used without revision upon application review."

3.34.1: Add:

"(k) Sea water regulating valves."

3.34.1.1, line 10: Delete "unit and power required" and substitute "unit, power, and motor current required".

PAGE 24

3.35: Delete and substitute:

"3.35 Manuals. - Manuals shall conform to type I of MIL-M-15071; photo views of the equipment shall be included as a part of the general description. A section shall be provided at the back of the manual containing reduced size copies of all drawings including diagram and assembly drawings of the air conditioning units, purge pump-out units, ancillary equipment, motor, electrical controllers, circuit breakers, and controls. The drawings may be referenced in order to amplify the test. In addition, those portions of the text which describe operation, maintenance procedures, assembly, disassembly, alinement, control adjustment, etc., shall be supplemented with sketches, sectional views, photographs, exploded views, or schematics, as appropriate, located as near as possible to the text. The text and amplifying material shall be sufficiently detailed to permit accomplishment of procedures or actions without repeated reference to drawings in the back of the manual."

PAGE 25

4.3, line 1: Delete "shockproof" and substitute "shock, proof".

4.3.3, second sentence: Delete and substitute: "The unit shall be tested class I or II of MIL-S-901, as specified (see 6.1)."

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PAGE 26

4.3.5, Delete and substitute:

"4.3.5 Capacity test for type I unit. A capacity test shall be conducted on each unit to determine if the specified capacity is reached. The unit shall be adjusted to deliver the specified chilled water temperature and chilled water flow and operated with at least 88°F condensing water and nominal 440-volt, 3-phase, 60-Hz power. The test also shall determine the ability of the unit to operate automatically at approximately 15, 50, and 100 percent of capacity. This shall be accomplished with chiller leaving water temperature at design conditions, and condenser water entering temperature and flow rate at design conditions, when operating under automatic controls. When testing the unit at 15 and 50 percent of rated capacity, the chilled water flow may be reduced to 50 percent of design conditions. Instrumentation shall have an accuracy of at least plus or minus 2 percent. Chilled water temperature shall be measured to within 1/10 of a degree F. The capacity test shall be run using chilled water flow as specified in 3.2.1. Failure to meet the specified tonnage capacity as a minimum, with calibrated instrumentation and corrected readings, shall be cause for rejection."

PAGE 27

4.4, item (a): Delete "(type II unit)."

4.4.1 and 4.4.1.1: Delete and substitute:

"4.4.1 Noise and vibration tests.

"4.4.1.1 Airborne. Airborne noise level tests shall be made on each type I and type II unit. The airborne noise testing shall be in accordance with the requirements, measurements, and data reporting procedures of MIL-STD-740. The units shall be noise tested in normal horizontal position while operating at approximately 15, 50, and 100 percent of rated capacity. The maximum allowable noise levels shall be as specified (see 6.1).

"4.4.1.2 Structureborne. Structureborne noise level tests shall be made on each type I unit, when specified (see 6.1), and each type II unit. The structureborne noise testing shall be in accordance with the requirements, measurements, and data reporting procedures of MIL-STD-740. The units shall be structureborne noise tested in normal horizontal position while operating at approximately 15, 50, and 100 percent of rated capacity. The maximum allowable structureborne noise levels shall be as specified (see 6.1).

"4.4.1.3 Vibration. Unless otherwise specified (see 6.1), one complete unit of each type, design, and size shall be vibration tested to determine conformance to the requirements specified in 3.13.3. The unit shall be vibration tested in normal horizontal position while operating at approximately 15, 50, and 100 percent of rated capacity.

"4.4.1.4 Prior to airborne noise, structureborne noise, or vibration testing, detailed procedures including sketches of the test set-up shall be submitted to NAVSEC for review.

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"4.4.1.5 For nonrotating, nonreciprocating machinery, if the first two units of a design are below the required limits by 10 db or more at all frequencies, following units on a contract or order need not be tested for noise unless otherwise specified (see 6.1)."

4.4.2.2: Add "All individual or cumulative leaks detected with the leak-detector set at the 1/2 ounce per year setting shall be repaired and retested to insure repair has been successful."

4.4.2.3 Delete and substitute:

"4.4.2.3 Prior to charging and operation testing, the entire refrigerant circuit shall be dehydrated and evacuated to a vacuum equivalent to 5 mm of mercury on an absolute pressure indicator. The introduction of additives such as anhydrous methyl alcohol to reduce freezing of residual moisture is not permitted. Upon evacuation of the unit to 5 mm mercury, the vacuum pump shall be secured and isolated from the unit. The evacuated unit shall stand for 6 hours. An increase in pressure, corrected for ambient temperature changes, of greater than 3 mm mercury shall be cause for rejection."

4.4.5: Add "An elapsed time log shall be included in the maintainability demonstration report to show the time required to disassemble and reassemble each component. Where maintenance on any component requires removal of another component(s), the log shall include the time required for disassembly and reassembly of all such components."

PAGES 28 thru 30

5. through 5.2.3: Delete and substitute:

"5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

"5.1 General requirements.

"5.1.1 First article sample. When specified (see 6.1) and equipment is required to be preserved-packaged level A and packaged level A or B, a first article pack of equipment, prior to package production, shall be subjected to the rough handling tests of MIL-P-116.

"5.1.2 Technical data. Complete descriptive details of the pack on drawing(s) shall be submitted to the procuring activity. Upon completion of first article testing, detailed test results shall be forwarded to the procuring activity. When specified in contract or order, the supplier shall submit packaging and transportation support data. All data shall be submitted to the procuring activity via the technical packaging element of the Administrating Contract Office (ACO).

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"5.1.2.1 Previous submittals. Data required by 5.1.2 and submitted on previously procured identical equipment is not required.

"5.1.3 Cleaning. Each equipment shall be cleaned so that all surfaces are free of grease, oil, flux, loose particles, and foreign materials. After completion of equipment testing, the oil, water, and refrigerant sides shall be drained and cleaned.

"5.1.3.1 Lubrication system. The lubrication system shall be flushed with normal operating lubricating oil drained, new filters installed, and recharged with new lubricating oil.

"5.1.3.2 Water side. The water side shall be blown dry with air that is free from oil, water, and dirt. Drains shall be opened to thoroughly remove trapped water.

"5.1.3.3 Refrigerant side. The refrigerant side of the condenser and cooler shall be cleaned before or after testing by pumping a cleaning solvent through the unit and a series of filters until no dirt or foreign particles remain in the system. The filters to be used shall be rated at 15 micrometre absolute (remove all particles larger than 15 micrometres). The pumping procedure shall be operated for a minimum of 20 minutes through clean filters after it has been indicated that no dirt or foreign particles remain. After equipment testing, the refrigerant side of the unit shall be dehydrated by evacuation to a vacuum of approximately 5 mm of Hg absolute pressure. The unit shall then be charged with dry, oil-free nitrogen to a positive gage pressure of approximately 5 lb/in^2. The system shall be provided with a separately mounted pressure gage or an installed system pressure gage may be used if suitable, for verification that the system is pressurized. In addition, a fitting/connection, if not currently installed shall be provided adjacent to the gage for system recharging, when required, during the time equipment is under storage. A tag shall be provided stating that the system is pressurized to 5 lb/in^2 with dry, oil-free nitrogen. In addition, the tag shall contain instructions for depressurization and repressurization of the system.

"5.1.4 Noise tested units (see 3.12). Units which have been noise tested shall be mounted and shipped on resilient mounts (see 3.13.4).

"5.1.5 Capillary tubes. Capillary tubes with bulbs shall be removed from their respective wells, coiled and tied, cushioned and wrapped and secured to the unit in a protected location. Each bulb package and well shall be tagged with each tag containing corresponding information for proper mating and reinstallation. Where used, a supply of heat conducting compound shall be provided for reinstallation of the thermal bulbs. Compound shall be in the quantity recommended by the equipment manufacturer and packaged in accordance with the manufacturer's retail procedure. Each package of compound shall contain complete identification markings, use instructions, storage recommendations and, if applicable, shelf life data.

"5.2 Preservation-packaging. Preservation-packaging shall be levels A and C as specified in 5.2.1.

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5.2.1 Levels A and C.

"5.2.1.1 Units. Each unit processed as specified in 5.1.3, 5.1.4 and 5.1.5 shall be unit protected to meet the requirements of method IIA of MIL-P-116. Exterior unpainted ferrous metal surfaces shall be coated with a preservative compound conforming to P-19 of MIL-P-116. Openings to the unit and attached components shall be covered to prevent entrance of foreign materials. Gages shall be cushioned and protected to prevent damage except the pressure-gage used for inspection of the pressurized refrigerant side. The flexible barrier bag shall be provided with an external mounted humidity indicator plug. Desiccant quantity shall be in accordance with MIL-P-116. The flexible barrier bag shall be fabricated from material conforming to MIL-B-22191, type I.

"5.2.1.2 Detached components. Detached accessories and parts shall be individually preserved-packaged level A in accordance with the applicable product specification. In the absence of such specifications, selection of the method of preservation-packaging shall be in accordance with MIL-P-116 and the intended use criteria therein. Packaged detached accessories and parts shall be consolidated within one or more of the following containers at the supplier's option:

Specification	Type or class
PPP-B-601	Domestic type
PPP-B-621	Class 1
PPP-B-636	Domestic
PPP-B-640	Class 1

Box closure shall be in accordance with the applicable box specification or appendix thereto; except that fiberboard boxes shall be closed with reinforced, pressure-sensitive tapes.

"5.2.1.2.1 Index list. Each container shall be provided with an index list itemizing the contents of the container. The list shall be placed in an opaque or transparent bag and closed. Closure shall be by heat sealing or the bag may be provided with a press or zip type closure. The list shall be placed inside the container on top of container contents to provide for ease of accessibility.

"5.3 Packing. Packing shall be level A, B, or C as specified in the contract or order.

"5.3.1 General.

"5.3.1.1 Center-of-gravity mounted unit(s). The equipment manufacturer shall provide a cradle/support structure for each unit(s) for support of the units(s) during transit and storage and which will remain with the unit(s) until the unit(s) shipboard installation. The support structure shall be marked REUSABLE - DO NOT DESTROY.

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"5.3.2 Levels A and B.

"5.3.2.1 Refrigerating unit(s). Each unit(s) shall be packed in a sheathed crate conforming to MIL-C-104, type II, class 2, and style at the supplier's option. Crates shall be provided with inspection door(s). The door(s) shall be located to permit visual reading of the refrigerant system pressure-gage and humidity indicator; and, of a size to permit access for repressurization of the system, if required. Crate closure shall be in accordance with the appendix to the crate specification.

"5.3.2.2 Detached accessories and parts. Detached accessories and parts, packaged as specified in 5.2.1.2, require no further packing. The container(s), where practicable shall be packed inside the refrigerating unit(s) container.

"5.3.2.3 Anchoring, blocking, bracing, and cushioning. Anchoring, blocking, bracing, and cushioning of the container contents shall be in accordance with the appendix to the crate specification and MIL-STD-1186.

"5.3.3 Level C.

"5.3.3.1 Unit(s). Unit(s), packaged as specified, shall be packed in containers acceptable to the common carrier and which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing, or method of shipment shall comply with the Uniform Freight or National Motor Freight Classification Rules or other carrier rules as applicable to the mode of transportation. Unit(s), when shipped uncrated by open type carrier, shall be covered with a tarpaulin or reinforced plastic covering to protect the equipment from damage.

"5.3.3.2 Detached accessories and parts. Detached accessories and parts shall be packed as specified in 5.3.3.1.

"5.4 Marking.

"5.4.1 Standard markings. In addition to any special markings required (see 6.1) or herein, interior and exterior containers and unpacked shipments shall be marked in accordance with MIL-STD-129. Level C marking requirements shall apply when level C, preparation for delivery, is specified in the contract or order.

"5.4.2 Special markings.

"5.4.2.1 Noise tested units. Each unit and its shipping container shall be tagged and marked 'SENSITIVE - NOISE TESTED UNIT - HANDLE WITH CARE'. Each packed unit(s) shall be provided with unpacking and handling instructions. These instructions shall be placed in a sealed waterproof envelope prominently marked 'UNPACKING AND HANDLING INSTRUCTIONS' and firmly affixed to the unit, if shipped unpacked, or to the outside of the shipping container in a protected location (preferably between the cleats on the end of the container adjacent to the identification marking). If the instructions cover a set of equipment packed in multiple containers, the instructions shall be affixed to the number one container of the set or system.

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"5.4.2.2 Refrigerant side (see 5.1.3.3). The required tag and instructions shall be protected as specified in 5.4.2.1. A surface adjacent to or the inspection door itself shall be marked 'REFRIGERANT SIDE PRESSURE GAGE - INSPECTION DOOR'.

"5.4.2.3 Structural markings. Structural markings of each unit(s) container is required and shall be in accordance with MIL-STD-129 and as specified herein. Containers specified in 5.3.2 shall be marked 'REUSABLE' including (a) center of balance, by means of a painted or stenciled black stripe approximately 1/2-inch wide on each side of the crate. The words 'CENTER OF BALANCE' in letters approximately 1-inch high shall be stenciled adjacent to the stripe on each side of the crate and, (b) sling points shall be legibly marked 'SLING HERE'.

"5.5 Unpacking instructions. Each crated unit(s) shall be provided with unpacking instructions. Instructions shall be protected as specified in 5.4.2.1.

"5.6 Depreservation instructions. Each unit(s) shall be provided a set of instructions covering the depreservation of the unit(s). Instructions shall state all information necessary for depreservation, assembly of detached accessories and parts, reinstallation of capillary tubes and other information necessary for assembly prior to installation/operation. Instructions shall be protected as specified in 5.4.2.1.

"5.7 Repair parts, tools, manuals and drawings.

"5.7.1 Repair parts and tools. Levels of protection for repair parts and tools shall be as follows:

Intended use	Preservation-packaging	Packing
Onboard	A	C
Stock	A	B
Immediate use	C	C

Except as otherwise specified herein, selection of the method of preservation and containers for packing shall be at the option of the equipment manufacturer and in accordance with MIL-R-196 and MIL-E-17555, as applicable. First article sample, rough handling test, and technical data specified in MIL-E-17555 are not required.

"5.7.1.1 Refrigerant. Refrigerant (see 3.3.1 and table I) shall be packaged, packed, and marked in accordance with Department of Transportation Rules and Regulations (DOT). In addition, shipment markings shall comply with MIL-STD-129 and color coding of MIL-STD-101.

"5.7.1.2 Lubricating oil. Lubricating oil (see table I) shall be furnished in one gallon, type V, class 4 or 6, level A packaging; packed level C, marked levels A and C respectively in accordance with MIL-STD-290.

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"5.7.2 Manuals.

"5.7.2.1 Accompanying equipment. Manuals shall be packaged in a grease-proof, waterproof plastic bag of minimum 4 mil thickness. Closure shall be accomplished by heat sealing or the bag may be provided with a zip or press fit type closure. Manuals shall be placed in the container housing the detached accessories and parts. The packing lists shall state which container contains the manuals. Containers shall also be marked including the approximate location therein.

"5.7.2.2 Bulk quantities. Manuals shall not be individually wrapped/packaged, but shall be packed and marked in accordance with the requirements of the applicable manual specification.

"5.7.3 Drawings and aperture cards. Drawings and aperture cards shall be prepared for shipment in accordance with the applicable item specification."

PAGES 30 and 31

6.1: Delete "(u) Whether units should be noise tested (see 4.4.1.5)."

6.1: Add:

"(x) When first article pack sample is required (see 5.1.1).
"(y) Special marking required (see 5.4.1)."

PAGE 31

Add as paragraph 6.5:

"6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped."

Preparing activity:
Navy - SH
(Project 4130-N231)

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